ETRI UWB FirmWare Programming Guide

Version 1.0

Shmt Co., Ltd.

REVISION HISTORY PAGE

Issue	Rev No.	Originator	Details of Change	Date
1	0.1	DJKIM/BHLEE SHKIM/BF109UP	Initial Version	2007-03-01
2	0.2	SHKIM/ BF109UP	Modify pre driver	2007-03-02
3	0.3	BHLEE	Update DMA	2007-03-06
4	0.4	DJKIM	Modify basic peripheral pre driver	2007-03-09
5	1.0	DJKIM/SHKIM /BF109UP	Add structure field comments	2007-03-26

Technical documents for Firmware

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1. INTRODUCTION

본 문서는 ETRI UWB의 firmware programming 시 사용 가능한 라이브러리 함수들에 대한 내용을 다룬다.

2. DRIVERS FUNCTIONS

ETRI UWB의 driver 함수는 pre/post-level driver 함수로 구성된다. Pre-level 드라이버 함수는 각 하드웨어 모듈의 레지스터 설정과 관련된 함수들이며, post-level 드라이버 함수들은 pre-level 드라이버를 이용한 각 하드웨어 모듈의 read/write와 같은 functional한 동작을 위한 드라이버 함수들이다.

3. BASIC PERIPHERALS

3.1. DMA

3.1.1. DMA2Init()

Description

DMA의 Schedule및 Resource를 초기화 한다.

시스템 시작과 동시에 한번만 실행을 해주면 된다.

```
void DMA2Init( smtBoolean polling)
```

Parameters

Polling

DMA동작 완료를 Polling으로 처리할지 인터럽트로 처리할지를 결정한다.

SMT_TRUE일 경우 Polling으로 처리하고 SMT_FALSE 일경우 인터럽트로 처리한다.

Return Value

none.

Remarks

none.

Requirement

Platform: Requires ETRI UWB firmware platform

Header: lib.h

Source: lib.c

Library: none

Example Code

```
none
```

3.1.2. smt2EDMACopy()

Description

Peripheral에서 memory로 DMA를 이용하여 데이터를 전송하는데 사용이 된다.

Memory에서 Memory로는 허용되지 않으며 그때는 smt2DMAMemCopy()를 사용해야 한다.

```
smtUint32 smt2EDMACopy(DMA_DEVICE device, EDMA_STRUCT *dma, smtUint8 descCount);
```

Parameters

device

EDMA에서는 미리정의된 peripheral에 따라 Channel이 정의되어 있다. 미리 정의된 device일 경우 해당하는 디바이스 값을 설정하고 그렇지 않을 경우 DMADEV_NONE를 사용한다.

Enumerator

Enumerator	Description
DMADEV_NAND	NAND device
DMADEV_SEIPTX / DMADEV_SEIPRX	SEIP device
DMADEV_I2STX / DMADEV_I2SRX	I2S device
DMADEV_SPI0TX / DMADEV_SPI0RX	SPI device
DMADEV_UART0TX / DMADEV_UART0RX DMADEV_UART1TX / DMADEV_UART1RX	UART 0/1 device
DMADEV_NONE	none

dma

EDMA 설정값이 있다. Src, dst를 지정하고 각각의 bus width및 한번에 전송할 블록등을 설정한다.

```
typedef struct
{
        smtUint32 src;                  // Source address
        smtUint32 dst;                  // Destination address
        smtUint32 nBytes;               // Transfer length

        smtBoolean m2m;                 // Memory to memory transfer flag
        DMA_BUSWIDTH srcWidth;          // Source bus width
        DMA_BUSWIDTH dstWidth;          // Destination bus width

        smtUint8 tSize;                 // Transfer size

        smtBoolean srcInc;              // Source address increment flag
        smtBoolean dstInc;              // Destination address increment flag
```

```
        smtBoolean startIntEn;          // Start interrupt enable flag
        smtBoolean EndIntEn;            // End interrupt enable flag

        ptDMAIntFunction intFunction;   // DMA interrupt function pointer
} EDMA_STRUCT;
```

descCount

디스크립터(DMA를 이용하여 전송하려는 내용을 여러 개를 디스크립터러 만들어서 한번에 적용하는데 사용)를 사용할경우 설정한 디스크립터 개수가 들어간다.

그렇지 않을경우 1 을 넣는다.(디스크립터를 사용하지 않음)

Return Value

none.

Remarks

none.

Requirement

Platform: Requires ETRI UWB firmware platform

Header: lib.h

Source: lib.c

Library: none

Example Code

```
EDMA_STRUCT edma;

        edma.src        = (smtUint32)&NANDDATA;
        edma.dst        = (smtUint32)ppagebuf;
        edma.nBytes     = sizeInByte;

        edma.m2m        = SMT_FALSE;
        edma.srcWidth   = transSize;
        edma.dstWidth   = DMA_BUSWIDTH32;
        edma.tSize      = burstSize;

        edma.srcInc = DMA_ADDR_NOINC;
        edma.dstInc = DMA_ADDR_INC;

        edma.startIntEn  = SMT_FALSE;
        edma.EndIntEn    = SMT_FALSE;

        edma.intFunction = &NANDDMAHandler;

        smt2EDMACopy(DMADEV_NAND, &descriptor, 1);
```

3.2. GPIO

3.2.1. smtGPIOSetDefault()

Description

GPIO0/GPIO1 port (32*2) default setting

```
void smtGPIOSetDefault(void);
```

Parameters

none

Return Value

none.

Remarks

Input mode, GPIO interrupt disable, Single active low edge interrupt

Requirement

Platform: Requires ETRI UWB firmware platform

Header: lib.h

Source: lib.c

Library: none

Example Code

```
none
```

3.2.2. smtGPIOSetDir()/smtGPIOGetDir()

Description

GPIO GPIO0_OE, GPIO1_OE register's software interface, set direction(in/out/bidirection) of GPIO

```
void smtGPIOSetDir (smtUint8 channel, smtUint8 gpioNum, smtUint32 mode);
void smtGPIOGetDir (smtUint8 channel, smtUint8 gpioNum, smtUint32 *mode);
```

Parameters

channel

GPIO port number

gpioNum

GPIO port bit number

mode

In/output mode

Return Value

none.

Remarks

none.

Requirement

Platform: Requires ETRI UWB firmware platform

Header: lib.h

Source: lib.c

Library: none

Example Code

```
none
```

3.2.3. smtGPIOSetData()/smtGPIOGetData()

Description

GPIO GPIO0_IN, GPIO0_OUT, GPIO1_IN, GPIO1_OUT register's software interface, write data or read data from GPIO controller

```
void smtGPIOSetData (smtUint8 channel, smtUint8 gpioNum, smtUint32 data);
void smtGPIOGetData (smtUint8 channel, smtUint8 gpioNum, smtUint32 *data);
```

Parameters

channel

GPIO port number

gpioNum

GPIO port bit number

data

data

Return Value

none.

Remarks

none.

Requirement

Platform: Requires ETRI UWB firmware platform

Header: lib.h

Source: lib.c

Library: none

Example Code

```
None
```

3.2.4. smtGPIOGetIntStatus ()

Description

GPIO port's gpio bit number interrupt status read

```
void smtGPIOGetIntStatus (smtUint8 channel, smtUint8 gpioNum, smtUint32 *gpioStatus);
```

Parameters

channel

GPIO port number

gpioNum

GPIO port bit number

gpioStatus

GPIO interrupt status of GPIO bit number

Return Value

none.

Remarks

none.

Requirement

Platform: Requires ETRI UWB firmware platform

Header: lib.h

Source: lib.c

Library: none

Example Code

```
None
```

3.2.5. smtGPIOIntClear()

Description

GPIO0_INTSTAT, GPIO1_INTSTAT register's software interface, clear GPIO interrupt pending register

```
void smtGPIOIntClear (smtUint8 channel, smtUint8 gpioNum);
```

Parameters

channel

GPIO port number

gpioNum

GPIO port bit number

Return Value

none.

Remarks

none.

Requirement

Platform: Requires ETRI UWB firmware platform

Header: lib.h

Source: lib.c

Library: none

Example Code

```
Refer to peri/gpio.c
```

3.2.6. smtGPIOIntEnable()

Description

GPIO0_INTEN, GPIO1_INTEN register's software interface, make enable/disable interrupt of GPIO

```
void smtGPIOIntEnable (smtUint8 channel, smtUint8 enable);
```

Parameters

channel

GPIO port

enable

GPIO port interrupt en/disable flag

Return Value

none.

Remarks

none.

Requirement

Platform: Requires ETRI UWB firmware platform

Header: lib.h

Source: lib.c

Library: none

Example Code

```
none
```

3.2.7. smtGPIOSetIntProperty()/smtGPIOGetIntProperty()

Description

GPIO0_INTLEVEL, GPIO0_INTPOL, GPIO0_INTBEDGE, GPIO0_INTLEVEL, GPIO0_INTPOL, GPIO0_INTBEDGE, register's software interface. Set GPIO's interrupt level, interrupt polarity, interrupt trigger mode.

```
void smtGPIOSetIntProperty (smtUint8 channel, GPIO_STRUCT gpio);
void smtGPIOGetIntProperty (smtUint8 channel, GPIO_STRUCT *gpio);
```

Parameters

channel

GPIO port

gpio

```
typedef struct
{
        smtUint32 intStatus;            // GPIO interrupt status
        smtUint32 intNum;               // GPIO interrupt number
        smtUint32 intLevel;             // GPIO interrupt level
        smtUint32 intPolarity;          // GPIO interrupt polarity
        smtUint32 intBothEdge;          // GPIO interrupt both edge

        smtUint8   gpioNum;             // GPIO number
} GPIO_STRUCT;
```

Return Value

none.

Remarks

none.

Requirement

Platform: Requires ETRI UWB firmware platform

Header: lib.h

Source: lib.c

Library: none

Example Code

```
none
```

3.3. TIMER

3.3.1. smtTIMERSetMode() / smtTIMERGetMode()

Description

Timer control register manipulate function

```
void smtTIMERSetMode (TIMER_STRUCT timer, TIMER_SEL timerSel);
void smtTIMERGetMode(TIMER_STRUCT *timer, TIMER_SEL timerSel);
```

Parameters

timer

```
            typedef struct
{
        smtUint32 data;                 // Timer data register value
        smtUint16 prescale;             // Timer prescaler value

        smtUint8 opMode;                // Timer operation mode
        smtBoolean timerClear;          // Timer counter register clear
        smtBoolean timerEn;             // Timer enable

} TIMER_STRUCT;
```

timerSel

Timer channel

Enumerator	Description
TIMER_SEL0	Timer channel 0
TIMER_SEL1	Timer channel 1
TIMER_SEL2	Timer channel 2
TIMER_SEL3	Timer channel 3

Return Value

none.

Remarks

none.

Requirement

Platform: Requires ETRI UWB firmware platform

Header: lib.h

Source: lib.c

Library: none

Example Code

Refer to peri/timerpwm.c

3.3.2. smtTIMERClrCount()

Description

Timer counter register clear function

```
void smtTIMERClrCount(smtUint8 timerCh);
```

Parameters

timerCh

Timer channel

Return Value

none.

Remarks

none.

Requirement

Platform: Requires ETRI UWB firmware platform

Header: lib.h

Source: lib.c

Library: none

Example Code

```
Refer to peri/timerpwm.c
```

3.3.3. smtTIMEROperation ()

Description

Timer operation(en/disable) function

```
void smtTIMEROperation (TIMER_STRUCT timer, TIMER_SEL timerSel);
```

Parameters

timer

```
        typedef struct
{
        smtUint32 data;                 // Timer data register value
        smtUint16 prescale;             // Timer prescaler value

        smtUint8 opMode;                // Timer operation mode
        smtBoolean timerClear;          // Timer counter register clear
        smtBoolean timerEn;             // Timer enable

} TIMER_STRUCT;
```

timerSel

Timer channel

Enumerator	Description
TIMER_SEL0	Timer channel 0
TIMER_SEL1	Timer channel 1
TIMER_SEL2	Timer channel 2
TIMER_SEL3	Timer channel 3

Return Value

none.

Remarks

none.

Requirement

Platform: Requires ETRI UWB firmware platform

Header: lib.h

Source: lib.c

Library: none

Example Code

```
none
```

3.4. VIC

3.4.1. EnableVIC()/DisableVIC()

Description

Write data or Read data form GPIO0/1

```
void EnableVIC(smtUint32 Polarity, smtUint32 Level, smtUint32 IntMod);
void DisableVIC(void);
```

Parameters

Polarity

Polarity of 32 interrupt source

Level

Interrupt event mode, level, edge mode

IntMod

Interrupt mode, IRQ/FIQ mode

Return Value

none.

Remarks

none.

Requirement

Platform: Requires ETRI UWB firmware platform

Header: irq.h

Source: irq.c

Library: none

Example Code

```
none
```

3.4.2. RequestIRQ()/ReleaseIRQ()

Description

Write data or Read data form GPIO0/1

```
void RequestIRQ(smtUint32 IRQ, ISRType ISR);
void ReleaseIRQ(smtUint32 IRQ);
```

Parameters

IRQ

IRQ number to Requested or Released

ISR

Interrupt handler to registry to VIC function table

Return Value

none.

Remarks

none.

Requirement

Platform: Requires ETRI UWB firmware platform

Header: irq.h

Source: irq.c

Library: none

Example Code

```
none
```

3.4.3. RequestIRQ()/ReleaseIRQ()

Description

Write data or Read data form GPIO0/1

```
void RequestIRQ(smtUint32 IRQ, ISRType ISR);
void ReleaseIRQ(smtUint32 IRQ);
```

Parameters

IRQ

IRQ number to Requested or Released

ISR

Interrupt handler to registry to VIC function table

Return Value

none.

Remarks

none.

Requirement

Platform: Requires ETRI UWB firmware platform

Header: irq.h

Source: irq.c

Library: none

Example Code

```
none
```

3.4.4. EnableIRQ()/DisableIRQ()

Description

Enable/Disable IRQ of VIC

```
void EnableIRQ(smtUint32 IRQ);
void DisableIRQ(smtUint32 IRQ);
```

Parameters

IRQ

IRQ number to enable or disable

Return Value

none.

Remarks

none.

Requirement

Platform: Requires ETRI UWB firmware platform

Header: irq.h

Source: irq.c

Library: none

Example Code

```
none
```

3.4.5. AckIRQ()

Description

Clear interrupt source pending register for edge interrupt

```
void AckIRQ(smtUint32 IRQ)
```

Parameters

IRQ

IRQ number to enable or disable

Return Value

none.

Remarks

none.

Requirement

Platform: Requires ETRI UWB firmware platform

Header: irq.h

Source: irq.c

Library: none

Example Code

```
none
```

3.4.6. InterruptHandler()

Description

System IRQ top interrupt handler

```
void InterruptHandler(void);
```

Parameters

none.

Return Value

none.

Remarks

none.

Requirement

Platform: Requires ETRI UWB firmware platform

Header: irq.h

Source: irq.c

Library: none

Example Code

```
none
```

3.4.7. UndefHandler()

Description

System undefined instruction handler

```
void UndefHandler(int IRQ);
```

Parameters

IRQ

IRQ number to enable or disable

Return Value

none.

Remarks

none.

Requirement

Platform: Requires ETRI UWB firmware platform

Header: irq.h

Source: irq.c

Library: none

Example Code

```
none
```

3.5. WDT

3.5.1. smtWDTSetMode () / smtWDTGetMode ()

Description

WDT control register manipulate function

```
void smtWDTSetMode(WDT_STRUCT wdt);
void smtWDTGetMode(WDT_STRUCT *wdt);
```

Parameters

wdt

```
typedef struct
{
        smtUint16 preScale;          // WDT prescaler register value
        smtUint16 reloadValue;       // WDT load register value

        smtUint8 div;                // WDT divide value
        smtBoolean clkSel;           // WDT clock select
        smtBoolean intEn;            // WDT interrupt en/disable
        smtBoolean rstEn;            // WDT reset en/disable
        smtBoolean wdtEn;            // WDT en/disable
} WDT_STRUCT;
```

Return Value

none.

Remarks

none.

Requirement

Platform: Requires ETRI UWB firmware platform

Header: lib.h

Source: lib.c

Library: none

Example Code

```
none
```

3.5.2. smtWDTGetCount ()

Description

WDT counter register read function

```
void smtWDTGetCount(smtUint16 *count);
```

Parameters

count

WDT counter register value

Return Value

none.

Remarks

none.

Requirement

Platform: Requires ETRI UWB firmware platform

Header: lib.h

Source: lib.c

Library: none

Example Code

```
none
```

3.5.3. smtWDTClearISR () / smtWDTGetISR ()

Description

WDT interrupt status register manipulate function

```
void smtWDTSetMode(void);
void smtWDTGetMode(smtUint8 *intrStatus);
```

Parameters

intrStatus

WDT interrupt status

Return Value

none.

Remarks

none.

Requirement

Platform: Requires ETRI UWB firmware platform

Header: lib.h

Source: lib.c

Library: none

Example Code

```
Refer to peri/wdt.c
```

4. COMMUNICATION DEVICES

4.1. I2C

4.1.1. smtI2CSetCtrl() / smtI2CGetCtrl()

Description

I2C 통신을 위한 clk, operation mode, interrupt/ack enable 등의 정보 set/get하는 함수

```
void smtI2CSetCtrl (I2C_CTRL_STRUCT i2cCtrl);
Void smtI2CGetCtrl (I2C_CTRL_STRUCT i2cCtrl);
```

Parameters

i2cCtrl

I2c 설정 구조체

```
typedef struct
{
        smtUint16 txPrescale;           // I2C prescale value
        smtUint8 intPendFlag;           // I2C interrupt pending flag
        smtUint8 opMode;                // I2C operation mode (Tx/Rx)
        smtUint8 ackEn;                 // I2C acknowledge enable
        smtUint8 txRxIntEn;             // I2C interrupt en/disable
} I2C_CTRL_STRUCT;
```

Return Value

void.

Remarks

none.

Requirement

Platform: Requires ETRI UWB firmware platform

Header: i2c_pre_drv.h

Source: i2c_pre_drv.c

Library: none

Example Code

```
None.
```

4.1.2. smtI2CGetStatus()

Description

이 함수는 I2C의 status 레지스터 정보를 가져온다.

```
void smtI2CGetStatus (I2C_STATUS_STRUCT i2cStat);
```

Parameters

i2cStat

I2c 설정 구조체

```
typedef struct
{
        smtBoolean outputEn;            // I2C output en/disable
        smtBoolean startStop;           // I2C start/stop
        smtBoolean busBusy;             // I2C bus busy/not busy
        smtBoolean arbitSta;            // I2C arbitration status
        smtBoolean ackValue;            // I2C receive ack status
        smtBoolean busError;            // I2C bus error status
} I2C_STATUS_STRUCT;
```

Return Value

void.

Remarks

none.

Requirement

Platform: Requires ETRI UWB firmware platform

Header: i2c_pre_drv.h

Source: i2c_pre_drv.c

Library: none

Example Code

```
None.
```

4.1.3. smtI2CSWReset()

Description

이 함수는 I2C soft reset한다.

```
void smtI2CSWReset (void);
```

Parameters

none

Return Value

void.

Remarks

none.

Requirement

Platform: Requires ETRI UWB firmware platform

Header: i2c_pre_drv.h

Source: i2c_pre_drv.c

Library: none

Example Code

```
None.
```

4.1.4. smtI2CEnable()

Description

이 함수는 I2C의 동작을 en/disable한다.

```
void smtI2CEnable (smtUint8 enable);
```

Parameters

enable

I2C en/disable flag

Return Value

void.

Remarks

none.

Requirement

Platform: Requires ETRI UWB firmware platform

Header: i2c_pre_drv.h

Source: i2c_pre_drv.c

Library: none

Example Code

```
None.
```

4.1.5. smtI2CInterruptWait()

Description

이 함수는 I2C의 pending flag가 발생하기를 기다린다.

```
void smtI2CInterruptWait (void);
```

Parameters

void.

Return Value

void.

Remarks

none.

Requirement

Platform: Requires ETRI UWB firmware platform

Header: i2c_pre_drv.h

Source: i2c_pre_drv.c

Library: none

Example Code

```
None.
```

4.1.6. smtI2CInterruptPendingClear()

Description

이 함수는 I2C의 pending flag를 clear 한다.

```
void smtI2CInterruptPendingClear (void);
```

Parameters

void

Return Value

void.

Remarks

none.

Requirement

Platform: Requires ETRI UWB firmware platform

Header: i2c_pre_drv.h

Source: i2c_pre_drv.c

Library: none

Example Code

```
None.
```

4.1.7. smtI2CInterruptPended()

Description

이 함수는 I2C pending flag를 return 한다.

```
smtInt32 smtI2CInterruptPended (void);
```

Parameters

void

Return Value

I2C pending status

Remarks

none.

Requirement

Platform: Requires ETRI UWB firmware platform

Header: i2c_pre_drv.h

Source: i2c_pre_drv.c

Library: none

Example Code

```
None.
```

4.1.8. smtI2CWrite()

Description

이 함수는 I2C를 통해 1byte 의 데이터 전송.

```
smtUint32smt2I2CWrite(
      smtUint8 addr,
      smtUint8 subAddr,
      smtUint8 data
);
```

Parameters

addr

slave device address

subAddr

slave device subaddress

data

전송할 데이터

Return Value

SMT_SUCCESS/ SMT_ERROR

Remarks

none.

Requirement

Platform: Requires ETRI UWB firmware platform

Header: i2c_pre_drv.h

Source: i2c_pre_drv.c

Library: none

Example Code

```
None.
```

4.1.9. smtI2CRead()

Description

이 함수는 I2C를 통해 slave device에서 데이터를 읽어들인다.

```
smtUint32smtI2CRead (
      smtUint8 addr,
      smtUint8 subAddr,
      smtUint8 *data
);
```

Parameters

addr

slave device address

subAddr

slave device subaddress

data

읽어들인 데이터

Return Value

SMT_SUCCESS/ SMT_ERROR

Remarks

none.

Requirement

Platform: Requires ETRI UWB firmware platform

Header: i2c_pre_drv.h

Source: i2c_pre_drv.c

Library: none

Example Code

```
None.
```

4.1.10. smtI2CInit()

Description

이 함수는 I2C를 초기화한다.

```
void smtI2CInit (smtUint16 prescaler);
```

Parameters

prescaler

I2C prescale value

Return Value

void

Remarks

none.

Requirement

Platform: Requires ETRI UWB firmware platform

Header: i2c_pre_drv.h

Source: i2c_pre_drv.c

Library: none

Example Code

```
None.
```

4.2. I2S

4.2.1. smtSetClockMode()/smtGetClockMode()

Description

I2S_CLKCTRL 의 software interface

```
void smtSetClockMode(I2S_CLOCK_MODE *pclkMode)
void smtGetClockMode(I2S_CLOCK_MODE *pclkMode)
```

Parameters

pclkMode

I2S_CLKCTRL register's 1:1 mapping structure

Return Value

none.

Remarks

none.

Requirement

Platform: Requires ETRI UWB firmware platform

Header: i2s_pre_drv.h

Source: i2s_pre_drv.c

Library: none

Example Code

```
Refer to test code (peri\comm\i2s.c)
```

4.2.2. smtI2SFIFOClear()

Description

I2S_STATUS RX/TX status clear bis's software interface

```
void smtI2SFIFOClear(I2S_TRANS_DIR dir)
```

Parameters

dir

I2S_DIR_RX is I2S RX FIFO clear, I2S_DIR_TX is I2S TX FIFO clear,

Return Value

none.

Remarks

none.

Requirement

Platform: Requires ETRI UWB firmware platform

Header: i2s_pre_drv.h

Source: i2s_pre_drv.c

Library: none

Example Code

```
Refer to test code (peri\comm\i2s.c)
```

none.

4.2.3. smtI2SFIFOReset()

Description

I2S_CTRL의 RX/TX reset bit software interface

```
void smtI2SFIFOReset(I2S_TRANS_DIR dir)
```

Parameters

dir

I2S_DIR_RX is I2S RX FIFO reset, I2S_DIR_TX is I2S TX FIFO reset,

Return Value

none.

Remarks

none.

Requirement

Platform: Requires ETRI UWB firmware platform

Header: i2s_pre_drv.h

Source: i2s_pre_drv.c

Library: none

Example Code

```
Refer to test code (peri\comm\i2s.c)
```

4.2.4. smtSetI2SMode()/smtGetI2SMode()

Description

I2S_CTRL's mode bits software interface

```
void smtSetI2SMode(I2S_TRANS_DIR dir, I2S_CONTROL *pcontrol)
```

Parameters

dir

I2S_DIR_RX is I2S RX FIFO reset, I2S_DIR_TX is I2S TX FIFO reset,

pcontrol

I2S_CTRL's DMAIntEn, FifoOURIntEn, WLen, Ljust, FifoTH 1:1 mapping structure

Return Value

none.

Remarks

none.

Requirement

Platform: Requires ETRI UWB firmware platform

Header: i2s_pre_drv.h

Source: i2s_pre_drv.c

Library: none

Example Code

```
Refer to test code (peri\comm\i2s.c)
```

4.2.5. smtI2SEnable()

Description

I2S_CTRL's RX/TX enable bit software interface

```
void smtI2SEnable(I2S_TRANS_DIR dir, smtUint32 enable)
```

Parameters

dir

Direction of I2S interface, I2S_DIR_RX is RX mode, I2S_DIR_TXs TX mode.

enable

true is enable, false is disable

Return Value

none.

Remarks

none.

Requirement

Platform: Requires ETRI UWB firmware platform

Header: i2s_pre_drv.h

Source: i2s_pre_drv.c

Library: none

Example Code

```
Refer to test code (peri\comm\i2s.c)
```

4.2.6. smtI2SGetStatus()

Description

I2S_STATUS's RX/TX status bit software interface

```
void smtI2SGetStatus(I2S_TRANS_DIR dir, I2S_STATUS_DAT *pstatus)
```

Parameters

dir

Direction of I2S interface, I2S_DIR_RX is RX mode, I2S_DIR_TXs TX mode.

pstatus

I2S_STATUS register 1:1 mapping layer per RX/TX

Return Value

none.

Remarks

none.

Requirement

Platform: Requires ETRI UWB firmware platform

Header: i2s_pre_drv.h

Source: i2s_pre_drv.c

Library: none

Example Code

```
Refer to test code (peri\comm\i2s.c)
```

4.2.7. smtI2SSetData()/smtI2SGetData()

Description

I2S_STATUS's RX/TX status bit software interface

```
void smtI2SSetData(smtUint32 data)
smtUint32 smtI2SGetData(void)
```

Parameters

data

write data to I2S_DATA register

Return Value

Read data from I2S_DATA register

Remarks

none.

Requirement

Platform: Requires ETRI UWB firmware platform

Header: i2s_pre_drv.h

Source: i2s_pre_drv.c

Library: none

Example Code

```
Refer to test code (peri\comm\i2s.c)
```

4.3. SPI

4.3.1. smtSPISetMode() / smtSPIGetMode()

Description

이 함수는 SPI의 모드를 set/get 한다.

```
void smtSPISetMode (smtUint8 channel, SPI_CON_STRUCT spiCtrl);
void smtSPIGetMode (smtUint8 channel, SPI_CON_STRUCT *spiCtrl);
```

Parameters

channel

SPI channel selection

spiCtrl

SPI control 구조체

```
typedef struct
{
        smtUint8 spiEnable;             // SPI en/disable
        smtUint8 masterSel;             // SPI master/slave select
        smtUint8 polaritySel;           // SPI clock polarity setting
        smtUint8 phaseSel;              // SPI clock phase setting
} SPI_CON_STRUCT;
```

Return Value

none.

Remarks

none.

Requirement

Platform: Requires ETRI UWB firmware platform

Header: spi_pre_drv.h

Source: spi_pre_drv.c

Library: none

Example Code

```
None.
```

4.3.2. smtSPISetPrescaler()

Description

이 함수는 SPI의 speed(prescale)를 설정한다.

```
void smtSPISetPrescaler (
      smtUint8 channel,
      smtUint32 preVal,
      smtUint32 divVal
);
```

Parameters

channel

SPI channel selection

preVal

SPI prescaler value

divVal

SPI divider value

Return Value

none.

Remarks

none.

Requirement

Platform: Requires ETRI UWB firmware platform

Header: spi_pre_drv.h

Source: spi_pre_drv.c

Library: none

Example Code

```
None.
```

4.3.3. smtSPIGetStatus()

Description

이 함수는 SPI의 상태 정보를 읽어온다.

```
void smtSPIGetStatus (smtUint8 channel, SPI_STATUS_STRUCT *spiStatus);
```

Parameters

channel

spi channel selection

spiStatus

SPI 상태 정보 구조체

```
typedef struct
{
        smtUint8 txFifoLevel;          // SPI Tx FIFO level
        smtUint8 rxFifoLevel;          // SPI Rx FIFO level
        smtUint8 txFifoFull;           // SPI Tx FIFO full flag
        smtUint8 txFifoEmpty;          // SPI Tx FIFO empty flag
        smtUint8 rxFifoFull;           // SPI Rx FIFO full flag
        smtUint8 rxFifoEmpty;          // SPI Rx FIFO empty flag
        smtUint8 busy;                 // SPI bus transfer status
} SPI_STATUS_STRUCT;
```

Return Value

none.

Remarks

none.

Requirement

Platform: Requires ETRI UWB firmware platform

Header: spi_pre_drv.h

Source: spi_pre_drv.c

Library: none

Example Code

```
None.
```

4.3.4. smtSPISetIntDMA() / smtSPIGetIntDMA()

Description

이 함수는 SPI의 interrupt en/disable 및 DMA en/disable 등을 set/get 한다.

```
void smtSPISetIntDMA (smtUint8 channel, SPI_INTDMA_STRUCT spiIntDMA);
void smtSPIGetIntDMA (smtUint8 channel, SPI_INTDMA_STRUCT *spiIntDMA);
```

Parameters

channel

SPI channel selection

spiIntDMA

SPI interrupt, DMA 설정 구조체

```
typedef struct
{
        smtUint8 txFifoIntEn;           // Tx FIFO interrupt enable
        smtUint8 rxFifoIntEn;           // Rx FIFO interrupt enable
        smtUint8 rxTimeOutIntEn;        // Rx timeout interrupt enable
        smtUint8 rxOverRunIntEn;        // Rx overrun interrupt enable
        smtUint8 txDMAEn;               // Tx DMA request enable
        smtUint8 txDMAlevel;            // Tx DMA request level
        smtUint8 rxDMAEn;               // Rx DMA request enable
        smtUint8 rxDMAlevel;            // Rx DMA request level
} SPI_INTDMA_STRUCT;
```

Return Value

none.

Remarks

none.

Requirement

Platform: Requires ETRI UWB firmware platform

Header: spi_pre_drv.h

Source: spi_pre_drv.c

Library: none

Example Code

```
None.
```

4.3.5. smtSPIGetIntStatus()

Description

이 함수는 SPI의 인터럽트 정보를 읽어온다.

```
void smtSPIGetIntStatus (smtUint8 channel, SPI_INTSTA_STRUCT *spitIntDMA);
```

Parameters

channel

SPI channel selection

spiIntDMA

SPI 인터럽트 상태 구조체

```
typedef struct
{
        smtUint8 rToutIntClr;           // Rx timeout interrupt clear
        smtUint8 rOverIntClr;           // Rx overrun interrupt clear
        smtUint8 txFifoIntSta;          // Tx FIFO interrupt status
        smtUint8 rxFifoIntSta;          // Rx FIFO interrupt status
        smtUint8 rxTimeOutIntSta;       // Rx timeout interrupt status
        smtUint8 rxOverRunIntSta;       // Rx overrun interrupt status
} SPI_INTSTA_STRUCT;
```

Return Value

none.

Remarks

none.

Requirement

Platform: Requires ETRI UWB firmware platform

Header: spi_pre_drv.h

Source: spi_pre_drv.c

Library: none

Example Code

```
None.
```

4.3.6. smtSPIClrIntStatus()

Description

이 함수는 SPI의 인터럽트 상태를 clear 한다.

```
void smtSPIClrIntStatus (smtUint8 channel, SPI_INTSTA_STRUCT spitIntDMA);
```

Parameters

channel

SPI channel selection

spiIntDMA

SPI 인터럽트 상태 구조체

```
typedef struct
{
        smtUint8 rToutIntClr;           // Rx timeout interrupt clear
        smtUint8 rOverIntClr;           // Rx overrun interrupt clear
        smtUint8 txFifoIntSta;          // Tx FIFO interrupt status
        smtUint8 rxFifoIntSta;          // Rx FIFO interrupt status
        smtUint8 rxTimeOutIntSta;       // Rx timeout interrupt status
        smtUint8 rxOverRunIntSta;       // Rx overrun interrupt status
} SPI_INTSTA_STRUCT;
```

Return Value

none.

Remarks

none.

Requirement

Platform: Requires ETRI UWB firmware platform

Header: spi_pre_drv.h

Source: spi_pre_drv.c

Library: none

Example Code

```
None.
```

4.3.7. smtSPIOnebyteWrite()

Description

이 함수는 SPI를 통해 1byte 전송한다.

```
void smtSPIOnebyteWrite (smtInt8 channel, smtInt8 data);
```

Parameters

channel

SPI channel selection

data

전송할 1 바이트 데이터

Return Value

none.

Remarks

none.

Requirement

Platform: Requires ETRI UWB firmware platform

Header: spi_pre_drv.h

Source: spi_pre_drv.c

Library: none

Example Code

```
None.
```

4.3.8. smtSPIOnebyteRead()

Description

이 함수는 SPI를 통해 1byte를 읽어온다.

```
smtUint8 smtSPIOnebyteRead (smtInt8 channel);
```

Parameters

channel

SPI channel selection

Return Value

smtUint8

읽어온 1 바이트 데이터

Remarks

none.

Requirement

Platform: Requires ETRI UWB firmware platform

Header: spi_pre_drv.h

Source: spi_pre_drv.c

Library: none

Example Code

```
None.
```

4.3.9. smtSPIInit()

Description

이 함수는 SPI를 통해 1byte 전송한다.

```
void smtSPIInit (
    smtInt8 channel,
    smtUint32 PreVal,
    smtUint32 DivVal
);
```

Parameters

channel

SPI channel selection

PreVal

SPI prescaler value.

DivVal

SPI divider value.

Return Value

none.

Remarks

none.

Requirement

Platform: Requires ETRI UWB firmware platform

Header: spi_pre_drv.h

Source: spi_pre_drv.c

Library: none

Example Code

```
None.
```

4.4. UART

4.4.1. smt2UartInit()

Description

UART 초기화.

```
smtBoolean
smt2UartInit(
    UartConfig uartCfg
);
```

Parameters

uartCfg

UART 초기화 정보를 담고 있는 구조체.

Return Value

SMT_SUCCESS 이면 수행 성공

Remarks

N/A

Requirement

Platform: Requires ETRI UWB firmware platform

Header: uart_post_drv.h

Source: uart_post_drv.c

Library: none

Example Code

```
none.
```

4.4.2. smt2UartDeInit ()

Description

initialize 한 UART 채널을 disable.

```
void
smt2UartDeInit(
    void
);
```

Parameters

none.

Return Value

none

Remarks

none

Requirement

Platform: Requires ETRI UWB firmware platform

Header: uart_post_drv.h

Source: uart_post_drv.c

Library: none

Example Code

```
none
```

4.4.3. smt2UartPutStr()

Description

UART를 통해 문자열을 출력.

```
smtBoolean
smt2UartPutStr(
    smtInt8 *pData
);
```

Parameters

pData

uart를 통해 출력할 문자열.

Return Value

SMT_SUCCESS 이면 수행 성공

Remarks

입력 문자열의 마지막은 '\0' 여야 한다.

Requirement

Platform: Requires ETRI UWB firmware platform

Header: uart_post_drv.h

Source: uart_post_drv.c

Library: none

Example Code

```
none.
```

4.4.4. smt2UartGetStr()

Description

UART를 통해 문자열을 입력 받는다.

```
smtBoolean
smt2UartGetStr(
    smtUint8 *pData,
    smtUint8 dataCnt
);
```

Parameters

pData

uart를 통해 출력할 문자열.

dataCnt

읽어들일 문자열의 개수.

Return Value

SMT_SUCCESS 이면 수행 성공

Remarks

none

Requirement

Platform: Requires ETRI UWB firmware platform

Header: uart_post_drv.h

Source: uart_post_drv.c

Library: none

Example Code

```
none.
```

4.4.5. smt2UartPrint()

Description

UART를 통해 포맷 형식의 문자열을 출력한다.

```
smtBoolean
smt2UartPrint(
    int dispLvl,
    char *fmt,
    [argument]...
);
```

Parameters

dispLvl

출력 레벨 설정값.

fmt

출력 포맷 설정.

[argument]…

선택적 인자들.

Return Value

SMT_SUCCESS 이면 수행 성공

Remarks

none

Requirement

Platform: Requires ETRI UWB firmware platform

Header: uart_post_drv.h

Source: uart_post_drv.c

Library: none

Example Code

```
none.
```

4.4.6. smt2UartPutCh()

Description

UART를 통해 한 개의 문자를 출력한다.

```
smtBoolean
smt2UartPutCh(
    smtInt8 data
);
```

Parameters

data

출력할 문자 데이터.

Return Value

SMT_SUCCESS 이면 수행 성공

Remarks

none

Requirement

Platform: Requires ETRI UWB firmware platform

Header: uart_post_drv.h

Source: uart_post_drv.c

Library: none

Example Code

```
none.
```

4.4.7. smt2UartGetCh()

Description

UART를 통해 한 개의 문자를 입력 받는다.

```
smtUint8
smt2UartGetCh(
    smtUint32 waitTime
);
```

Parameters

waitTime

time out value.

Return Value

SMT_SUCCESS 이면 수행 성공

Remarks

none

Requirement

Platform: Requires ETRI UWB firmware platform

Header: uart_post_drv.h

Source: uart_post_drv.c

Library: none

Example Code

```
none.
```

4.4.8. smtUartSetMode()

Description

UART master/status/baud generation/RX timeout register 변경.

```
smtBoolean
smtUartSetMode(
    UartChannel uartCh,
    UartStruct *pUart,
    UartMode mode
);
```

Parameters

uartCh

UART channel 선택.

pUart

register setting 정보를 담고 있는 구조체 포인터.

mode

setting 할 target register를 선택.

Return Value

SMT_SUCCESS 이면 수행 성공

Remarks

N/A

Requirement

Platform: Requires ETRI UWB firmware platform

Header: uart_pre_drv.h

Source: uart_pre_drv.c

Library: none

Example Code

```
none.
```

4.4.9. smtUartGetMode()

Description

UART master/status/baud generation/RX timeout register 값을 읽어서 인수로 넘어온 구조체에 저장한다..

```
smtBoolean
smtUartGetMode(
    UartChannel uartCh,
    UartStruct *pUart,
    UartMode mode
);
```

Parameters

uartCh

UART channel 선택.

pUart

register setting 정보를 저장할 구조체 포인터.

mode

읽어들일 target register를 선택.

Return Value

SMT_SUCCESS 이면 수행 성공

Remarks

N/A

Requirement

Platform: Requires ETRI UWB firmware platform

Header: uart_pre_drv.h

Source: uart_pre_drv.c

Library: none

Example Code

```
none.
```

4.4.10. smtUartReadRxFifo()

Description

UART RX FIFO에서 1byte size의 데이터를 읽어 온다.

```
smtUint8
smtUartReadRxFifo(
    UartChannel uartCh
);
```

Parameters

uartCh

UART channel 선택.

Return Value

RX FIFO에 읽어 들인 1byte size의 데이터.

Remarks

N/A

Requirement

Platform: Requires ETRI UWB firmware platform

Header: uart_pre_drv.h

Source: uart_pre_drv.c

Library: none

Example Code

```
none.
```

4.4.11. smtUartWriteTxFifo()

Description

UART RX FIFO에서 1byte size의 데이터를 읽어 온다.

```
void
smtUartWriteTxFifo(
    UartChannel uartCh,
    smtUint8 txData
);
```

Parameters

uartCh

UART channel 선택.

txData

TX FIFO에 쓸 데이터.

Return Value

none.

Remarks

N/A

Requirement

Platform: Requires ETRI UWB firmware platform

Header: uart_pre_drv.h

Source: uart_pre_drv.c

Library: none

Example Code

```
none.
```

4.4.12. smtUartDataAvailable()

Description

UART RX FIFO의 count 값을 체크 하여 읽어 들일 데이터가 있는지를 검사한다.

```
smtUint32 smtUartDataAvailable(void);
```

Parameters

none.

Return Value

none.

Remarks

N/A

Requirement

Platform: Requires ETRI UWB firmware platform

Header: uart_pre_drv.h

Source: uart_pre_drv.c

Library: none

Example Code

```
none.
```

5. STORAGE DEVICES

5.1. NAND

5.1.1. smtNANDSetOPT0()

Description

NAND 1st Operation's software interface

```
void smtNANDSetOPT0(NAND_OPT0 *popt0);
```

Parameters

popt0

NAND 1st Operation's 1:1 mapping

Return Value

none.

Remarks

none.

Requirement

Platform: Requires ETRI UWB firmware platform

Header: nand_pre_drv.h

Source: nand _pre_drv.c

Library: none

Example Code

```
Refer to test code (peri\storage\nand.c)
```

5.1.2. smtNANDSetOPT1()

Description

NAND 2st Operation's software interface

```
void smtNANDSetOPT1(NAND_OPT1 *popt1);
```

Parameters

popt1

NAND 1st Operation's 1:1 mapping

Return Value

none.

Remarks

none.

Requirement

Platform: Requires ETRI UWB firmware platform

Header: nand_pre_drv.h

Source: nand _pre_drv.c

Library: none

Example Code

```
Refer to test code (peri\storage\nand.c)
```

5.1.3. smtNANDSetOPT2()

Description

NAND 3st Operation's software interface

```
void smtNANDSetOPT1(NAND_OPT2 *popt2);
```

Parameters

popt1

NAND 2st Operation's 1:1 mapping

Return Value

none.

Remarks

none.

Requirement

Platform: Requires ETRI UWB firmware platform

Header: nand_pre_drv.h

Source: nand _pre_drv.c

Library: none

Example Code

```
Refer to test code (peri\storage\nand.c)
```

5.1.4. smtNANDGetData()/smtNANDSetData()

Description

NAND, NFDATA register software interface

```
void smtNANDSetData(smtUint32 *pdata)
smtUint32 smtUint32 smtNANDGetData(void)
```

Parameters

pdata

data for write NAND

Return Value

Data for read from NAND.

Remarks

none.

Requirement

Platform: Requires ETRI UWB firmware platform

Header: nand_pre_drv.h

Source: nand _pre_drv.c

Library: none

Example Code

```
Refer to test code (peri\storage\nand.c)
```

5.1.5. smtNANDSetCfg()/smtNANDGetCfg()

Description

NAND, NFCONF register software interface

```
void smtNANDSetCfg(NAND_CFG *pconfig)
void smtNANDGetCfg(NAND_CFG *pconfig)
```

Parameters

pconfig

NAND, NFCONF register 1:1 mapping structure

Return Value

none

Remarks

none.

Requirement

Platform: Requires ETRI UWB firmware platform

Header: nand_pre_drv.h

Source: nand _pre_drv.c

Library: none

Example Code

```
Refer to test code (peri\storage\nand.c)
```

5.1.6. smtNANDSetControl()/smtNANDGetControl()

Description

NAND, NFCTRL register software interface

```
void smtNANDSetControl(NAND_CTRL *pcontrol)
void smtNANDGetControl(NAND_CTRL *pcontrol)
```

Parameters

pcontrol

NAND, NFCTRL register 1:1 mapping structure

Return Value

none

Remarks

none.

Requirement

Platform: Requires ETRI UWB firmware platform

Header: nand_pre_drv.h

Source: nand _pre_drv.c

Library: none

Example Code

```
Refer to test code (peri\storage\nand.c)
```

5.1.7. smtNANDSetStatus()/smtNANDGetStatus()

Description

NAND, NFSTAT register software interface

```
void smtNANDSetStatus(NAND_STAT *pstatus)
void smtNANDGetStatus(NAND_STAT *pstatus)
```

Parameters

pstatus

NAND, NFSTAT register 1:1 mapping structure

Return Value

none

Remarks

none.

Requirement

Platform: Requires ETRI UWB firmware platform

Header: nand_pre_drv.h

Source: nand _pre_drv.c

Library: none

Example Code

```
Refer to test code (peri\storage\nand.c)
```

5.1.8. smtNANDGetFIFOStatus()

Description

NAND, NFFIFOSTAT register software interface

```
void smtNANDGetFIFOStatus(NAND_FIFOSTAT *pfifoStatus)
```

Parameters

pstatus

NAND, NFFIFOSTAT register 1:1 mapping structure

Return Value

none

Remarks

none.

Requirement

Platform: Requires ETRI UWB firmware platform

Header: nand_pre_drv.h

Source: nand _pre_drv.c

Library: none

Example Code

```
Refer to test code (peri\storage\nand.c)
```

5.1.9. smtNANDGetECC()

Description

NAND, ECCSECTOR0~ECCSECTOR8, SECCSECTOR0~SECCSECTOR3 register software interface

```
void smtNANDGetECC(smtUint8 eccIdx, smtUint32 *pecc, smtUint32 *psecc)
```

Parameters

eccIdx

wanted Main/Spare ECC number

pecc

smtUint32 pointer for Main ECC, if NULL no operation for this pointer

psecc

smtUint32 pointer for Spare ECC, if NULL no operation for this pointer

Return Value

none

Remarks

none.

Requirement

Platform: Requires ETRI UWB firmware platform

Header: nand_pre_drv.h

Source: nand _pre_drv.c

Library: none

Example Code

```
Refer to test code (peri\storage\nand.c)
```

5.2. SRAM

5.2.1. smtSMCGetControl()/smtSMCSetControl()

Description

ESMC_B0_CON, ESMC_B1_CON, ESMC_B2_CON, ESMC_B3_CON register software interface

```
void smtSMCGetControl(smtUint8 bank, SMC_CTRL *pcontrol)
void smtSMCSetControl(smtUint8 bank, SMC_CTRL *pcontrol)
```

Parameters

bank

ESMC_B0_CON, ESMC_B1_CON, ESMC_B2_CON, ESMC_B3_CON register selector, i.e. 0 means ESMC_B0_CON.

pcontrol

ESMC_B[0-3]_CON register's 1:1 mapping layer

Return Value

none.

Remarks

none.

Requirement

Platform: Requires ETRI UWB firmware platform

Header: smc_pre_drv.h

Source: smc_pre_drv.c

Library: none

Example Code

```
none
```

5.2.2. smtSMCGetControl()/smtSMCSetControl()

Description

ESMC_B0_CON, ESMC_B1_CON, ESMC_B2_CON, ESMC_B3_CON register software interface

```
void smtSMCGetControl(smtUint8 bank, SMC_CTRL *pcontrol)
void smtSMCSetControl(smtUint8 bank, SMC_CTRL *pcontrol)
```

Parameters

bank

ESMC_B0_CON, ESMC_B1_CON, ESMC_B2_CON, ESMC_B3_CON register selector, i.e. 0 means ESMC_B0_CON.

pcontrol

ESMC_B[0-3]_CON register's 1:1 mapping layer

Return Value

none.

Remarks

none.

Requirement

Platform: Requires ETRI UWB firmware platform

Header: smc_pre_drv.h

Source: smc_pre_drv.c

Library: none

Example Code

```
none
```

5.3. DDR-SDRAM

5.3.1. smtDMCSetControl()/smtDMCGetControl()

Description

DDRCON register software interface

```
void smtDMCSetControl(DMC_CTRL *pcontrol)
void smtDMCGetControl(DMC_CTRL *pcontrol)
```

Parameters

pcontrol

DDRCON register’s 1:1 mapping structure

Return Value

none.

Remarks

none.

Requirement

Platform: Requires ETRI UWB firmware platform

Header: dmc_pre_drv.h

Source: dmc_pre_drv.c

Library: none

Example Code

```
none
```

5.3.2. smtDMCGetTimingControl()/smtDMCSetTimingControl()

Description

DDRTCON register software interface

```
void smtDMCSetTimingControl(DMC_TIMING *ptiming)
void smtDMCGetTimingControl(DMC_TIMING *ptiming)
```

Parameters

ptiming

DDRTCON register's 1:1 mapping structure

Return Value

none.

Remarks

none.

Requirement

Platform: Requires ETRI UWB firmware platform

Header: dmc_pre_drv.h

Source: dmc_pre_drv.c

Library: none

Example Code

```
none
```

5.3.3. smtDMCGetPowerControl()/smtDMCSetPowerControl()

Description

DDRPCON register software interface

```
void smtDMCSetPowerControl(DMC_POWER *ppower)
void smtDMCGetPowerControl(DMC_POWER *ppower)
```

Parameters

ppower

DDRPCON register's 1:1 mapping structure

Return Value

none.

Remarks

none.

Requirement

Platform: Requires ETRI UWB firmware platform

Header: dmc_pre_drv.h

Source: dmc_pre_drv.c

Library: none

Example Code

```
none
```

5.3.4. smtDMCGetRefreshControl()/smtDMCSetRefreshControl()

Description

DDRREF register software interface

```
void smtDMCSetRefreshControl(DMC_REFRESH *prefresh)
void smtDMCGetRefreshControl(DMC_REFRESH *prefresh)
```

Parameters

prefresh

DDRREF register's 1:1 mapping structure

Return Value

none.

Remarks

none.

Requirement

Platform: Requires ETRI UWB firmware platform

Header: dmc_pre_drv.h

Source: dmc_pre_drv.c

Library: none

Example Code

```
none
```